Module 1 - Linux Infrastructure Baseline and Recovery

Project: Three-VM, two-tier e-commerce homelab

Platform: CentOS Stream 10, VirtualBox

Repository: `my-devops-journey`

Outcome: Baseline captured, recovery snapshots created, cloned machine IDs corrected, and application connectivity validated.

1. Project objective

Establish a reproducible baseline of a three-VM Linux environment before further DevOps exercises. Document compute, memory, storage, packages, services, networking, routing, firewall, recovery checkpoints, and system identity. Validate that a maintenance change does not disrupt the deployed e-commerce application.

2. Architecture

```
Windows host (host-only network 192.168.56.0/24)
  |-- lab-server-1: 192.168.56.10 (web)
  |-- lab-server-2: 192.168.56.20 (database)
  `-- lab-router:   192.168.56.30 (router)

lab-server-1                   lab-router                    lab-server-2
192.168.10.10/24  ---->  192.168.10.1 | 192.168.20.1  ----> 192.168.20.10/24
Apache + PHP-FPM             IPv4 forwarding/firewalld        MariaDB :3306
                                    |
                             NAT 10.0.5.15/24
                             gateway 10.0.5.2
```

The web and database VMs have default gateways `192.168.10.1` and `192.168.20.1`, respectively. Router interfaces `enp0s3` and `enp0s8` serve these subnets; `enp0s9` serves host-only management; `enp0s10` serves NAT.

3. Baseline inventory

Item	lab-server-1 (web)	lab-server-2 (database)	lab-router
OS	CentOS Stream 10	CentOS Stream 10	CentOS Stream 10 (same cloned layout; OS version not separately printed in this session)
vCPU	2	2	2
RAM	3.5 GiB	3.5 GiB	3.5 GiB
Disk	500 GB	500 GB	500 GB
Root volume	70 GB XFS	70 GB XFS	70 GB XFS
`/home` volume	~424 GB XFS	~424 GB XFS	~424 GB XFS
Swap volume	~4.8 GB	~4.8 GB	~4.8 GB
Root used at inspection	5%	5%	5%
Primary workload	Apache, PHP-FPM	MariaDB	Linux routing and NAT
Recovery snapshot	`module-1-baseline-web`	`module-1-baseline-db`	`module-1-baseline-router`

The web and database hosts reported kernel `6.12.0-43.el10.x86_64`. The router's kernel was not separately checked in the captured outputs.

LVM findings

On `lab-server-1`, `pvs`, `vgs`, and `lvs` confirmed one physical volume `/dev/sda3`, one volume group `cs`, and three logical volumes: `root` (70.00 GB), `home` (424.18 GB), and `swap` (4.81 GB). `PFree=0` and `VFree=0`: no unallocated LVM extents, despite substantial free space **inside** existing filesystems. `/var/www/html` resides on the root filesystem, not `/home`.

4. Software and service baseline

Web host (lab-server-1):

- Apache HTTP Server `httpd-2.4.63-17.el10.x86_64`
- PHP and PHP-FPM `8.3.33-1.el10.x86_64`
- PHP MySQL driver `php-mysqlnd-8.3.33-1.el10.x86_64`
- MariaDB **client** `10.11.18-1.el10.x86_64`
- Git `2.52.0-1.el10.x86_64`
- firewalld `2.3.0-2.el10.noarch`
- 21 running systemd service units, including `httpd`, `php-fpm`, `sshd`, `firewalld`, and `NetworkManager`.

Database host (lab-server-2):

- MariaDB server and client `10.11.18-1.el10.x86_64`
- firewalld `2.3.0-2.el10.noarch`
- 20 running systemd service units, including `mariadb`, `sshd`, `firewalld`, and `NetworkManager`.
- `systemctl is-active mariadb` returned `active` and `systemctl is-enabled mariadb` returned `enabled`, including after reboot.

Installed packages and running services were checked separately: package installation does not imply service availability.

5. Networking and router configuration

VM	Interface	Address	Purpose
lab-server-1	enp0s3	192.168.10.10/24	Web subnet
lab-server-1	enp0s9	192.168.56.10/24	Management
lab-server-2	enp0s3	192.168.20.10/24	Database subnet
lab-server-2	enp0s9	192.168.56.20/24	Management
lab-router	enp0s3	192.168.10.1/24	Web gateway
lab-router	enp0s8	192.168.20.1/24	Database gateway
lab-router	enp0s9	192.168.56.30/24	Management
lab-router	enp0s10	10.0.5.15/24	NAT, DHCP

Router default route: `default via 10.0.5.2 dev enp0s10`. Server default routes point to their respective router subnet IPs. The web host's `/etc/resolv.conf` contained `1.1.1.3` and `1.0.0.3`, generated by NetworkManager; DNS reachability was not separately tested.

Router settings observed:

- `net.ipv4.ip_forward = 1`, configured persistently in `/etc/sysctl.d/99-lab-router.conf`.
- firewalld `internal` zone: `enp0s3`, `enp0s8`; `forward: yes`.
- firewalld `external` zone: `enp0s10`; `masquerade: yes`.
- firewalld `public` zone: `enp0s9`.
- Active `lab-internet` policy: ingress `internal`, egress `external`, target ACCEPT.
- NAT DHCP lease observed as `10.0.5.15`, gateway and DHCP server `10.0.5.2`; DHCP client identifier `01:08:00:27:08:b3:99`.

The router's active firewall policy and zones were verified again after reboot. The complete permanent firewalld configuration was not separately audited in this module.

6. Recovery snapshots

Before system-identity maintenance, VirtualBox snapshots were created:

VM	Snapshot
lab-server-1	`module-1-baseline-web`
lab-server-2	`module-1-baseline-db`
lab-router	`module-1-baseline-router`

Important: A VirtualBox snapshot is a rollback checkpoint, not a substitute for an independent backup. Snapshot consistency and restore testing were not independently demonstrated in this exercise.

7. Incident: duplicate cloned machine IDs

Symptom and investigation

All three VMs initially returned the same `/etc/machine-id`:

```
c2f45dea8c95415f8c2d3656407e79ef
```

This was consistent with cloned installations inheriting the original identity. The web VM retained the original ID; the database and router VMs received new IDs.

Risk controls

1. Created recovery snapshots before modification.
2. Verified NetworkManager profiles and addressing before changing identities.
3. Confirmed both database interfaces used `ipv4.method: manual`.
4. Confirmed the router's web, database, and management interfaces used `manual`; NAT used DHCP.
5. Checked router DHCP lease and client identifier.
6. Changed **one VM at a time** and rebooted before moving to the next.

Commands used on each VM needing a new identity

```
sudo truncate -s 0 /etc/machine-id
sudo systemd-machine-id-setup
cat /etc/machine-id
sudo reboot
```

Caution: These commands modify a core system identity and should not be run casually on a production host. `systemd-machine-id-setup` can obtain identity from existing sources, so the result must be checked for uniqueness. Plan maintenance, understand application dependencies, and have a tested recovery strategy before use.

Final machine IDs

VM	Final machine ID
lab-server-1	`c2f45dea8c95415f8c2d3656407e79ef`
lab-server-2	`dbf74d72345f47b9ab5dcafdb8e5944e`
lab-router	`fb8fae4830834466b8e8f625e7f4a9c8`

On `lab-server-2`, `/var/lib/dbus/machine-id` did not exist; no separate D-Bus identity file was modified. After reboot, database IPs/routes and MariaDB `active/enabled` status were unchanged. After router reboot, hostname, four interface addresses, default route, forwarding setting, active zones, and `lab-internet` policy were intact.

8. Post-change validation evidence

All tests below were run **from lab-server-1 after both modified VMs rebooted**.

Cross-subnet connectivity

```
ping -c 4 192.168.20.10
```

Result: **4 transmitted, 4 received, 0% packet loss**, average RTT **2.559 ms**.

Database TCP reachability

```
timeout 5 bash -c 'echo > /dev/tcp/192.168.20.10/3306' \
  && echo 'Database port reachable' \
  || echo 'Database port unreachable'
```

Result: `Database port reachable`. This verifies TCP connectivity, **not** database authentication or SQL execution.

Application HTTP response

```
curl -sS -o /tmp/ecommerce-check.html \
  -w 'HTTP status: %{http_code}\n' http://localhost/
grep -Ei 'Drone|Laptop|Tablet|Watch' /tmp/ecommerce-check.html | head -n 8
```

Result: **HTTP 200** and product HTML including Laptop (`100$`), Drone (`200$`), and Tablet (`50$`). This supports application health; no direct SQL query was run in this final check.

9. Useful command reference

```
# Host resources
lscpu
free -h
uptime
lsblk -o NAME,SIZE,FSTYPE,TYPE,MOUNTPOINTS
df -hT
df -i

# LVM inspection
sudo pvs
sudo vgs
sudo lvs

# OS, packages, services
cat /etc/os-release
uname -r
rpm -q httpd php php-fpm php-mysqlnd mariadb git firewalld
systemctl list-units --type=service --state=running --no-pager
systemctl is-active mariadb
systemctl is-enabled mariadb

# Networking
ip -br -4 addr
ip -4 route
nmcli -f NAME,UUID,TYPE,DEVICE connection show
sudo firewall-cmd --get-active-zones
sudo firewall-cmd --list-all-policies
sysctl net.ipv4.ip_forward

# System identity
hostnamectl --static
cat /etc/machine-id
```

10. Interview talking points

Q: What is the difference between free filesystem space and free LVM space?

A: `df -hT` reports unused capacity inside a mounted filesystem. `vgs` reports unallocated extents in a volume group. A volume group can have `VFree=0` while its existing filesystems have ample unused capacity.

Q: How do you verify that a Linux service survives reboot?

A: Check `systemctl is-enabled`, reboot under a controlled change, then check `systemctl is-active` and perform an application-level health test.

Q: Why did you change the cloned machine IDs?

A: Each independent Linux installation should have a distinct identity. The cloned VMs shared the same ID, so I created snapshots, checked network dependencies, changed one identity at a time, rebooted, and validated the services and network.

Q: Why is ping insufficient to prove an application works?

A: Ping confirms ICMP reachability only. I also tested TCP port 3306 and checked that Apache returned HTTP 200 with product content.

Q: What does a default route do?

A: It provides a next hop for destinations not covered by a more specific route. Here the web host sends database-subnet traffic to `192.168.10.1`, and the database host returns traffic via `192.168.20.1`.

11. Follow-up checks / limitations

- Perform a direct authenticated SQL query from the web tier if deeper DB verification is required.
- Audit firewalld permanent configuration explicitly and verify persistence after further maintenance.
- Validate a snapshot restoration procedure in a disposable clone rather than risking the working environment.
- Consider an independent MariaDB backup, especially before later filesystem or database exercises.
- Record router kernel/package baseline if a complete per-host software inventory is required.
- Check DNS resolution independently if name-based connectivity becomes part of the lab.

12. Outcome

This module established a baseline across a three-VM CentOS Stream lab, corrected duplicate cloned system identities, and verified network and application functionality after reboot. It demonstrates Linux administration, LVM inspection, NetworkManager troubleshooting, systemd service validation, firewall/routing analysis, recovery planning, and controlled change management.